

### Broader interests - Hydrodynamical modeling of protoplanetary disks

Planets form in protoplanetary disks [1]. Disk observations from ALMA, VLT(I), etc., today are able to constrain the dynamical and chemical conditions under which small dust can grow into planets [2,3]. My research focuses on the **theoretical modeling of protoplanetary disks**, with a particular emphasis on **radiation hydrodynamics, dust dynamics, and hydrodynamical instabilities**—laying out the foundations to interpret observations. Across seven projects—three as lead author, two as second author, and two as a collaborator—I am investigating the physical processes that shape the early evolution of planetary systems. These range from binary disk dynamics and irradiated disk geometries to gravitational and streaming instabilities. My work combines analytical insights with high-resolution simulations using state-of-the-art codes such as PLUTO [4,5,6].

My research on protoplanetary disks began during my Master's studies at the University of Tübingen. Prior to that, I was interested in physics, applied mathematics, and statistics, with some experience in astrophysics—photometry and orbital dynamics of exoplanets. During my PhD at the Max Planck Institute for Astronomy, I have continued to develop disk models and have established collaborations with researchers across Europe, the UK, the US, and India. The following is a chronological summary of my research contributions to this field.

**Main project #1: How cooling influences circumbinary disks** [Sudarshan et al. 2022, ADS] Disks around binary stars show large, eccentric cavities [7] whose properties depend on the disk's aspect ratio, viscosity, binary eccentricity and mass ratio [8,9]. Recent studies show that radiative cooling significantly affects the shape and size of gaps/cavities created by planets [10,11]. We studied the role of different thermodynamical models—locally isothermal, radiatively cooled, and parameterized  $\beta$ -cooling [12]—on the cavity properties of circumbinary disks using 2D hydrodynamical simulations for varying binary eccentricities, and over very long timescales. While radiative and locally isothermal models yield comparable cavity shapes, the inner disk structure and precession rates differ. With  $\beta$ -cooled models, the shape and size of the cavity changes dramatically towards values of  $\beta = 1$  (see Fig. 1a). We propose a parameterized  $\beta$  model that captures the shorter cooling timescales within the cavity, closely reproducing radiative model results and emphasizing the importance of accurate thermodynamical treatment.

**Related collaboration #1: HD45677: the inner cavity of a possible close binary disk** [Gravity Collaboration, Sanchez-Bermudez et al. (incl. Sudarshan) in prep.] New GRAVITY and PIONIER reductions for HD 45677 [13,14,15], combined with Br $\gamma$  imaging, reveal orbital motion of a dusty feature and high-velocity deviations indicative of a potential secondary companion. Our hydrodynamical models (see Fig. 1b) and mock observations constrain the cavity temperature and place an upper limit on the binary mass ratio, offering insights into the system's dynamics.

**Master thesis, project #2: Spirals and gravitoturbulent disks in 2D** [Sudarshan et al. in prep., thesis] A massive, gravitationally unstable disk settles into a system of stable angular momentum transport by generating spiral arms under slow cooling [16,17]. The underlying dynamics induced by the gravitational instability (GI) determine the observable disk morphology and small-scale turbulence, and can affect the efficiency of planet formation from fragmentation scenarios [18]. We perform 2D global numerical simulations of self-gravitating accretion disks using the finite-volume code PLUTO in cylindrical geometry with a parameterized cooling law. We characterize and explain the disk structure, and compare it to previous 3D analyses [19] consisting of localized spirals. We find that the spirals show self-similarity and orient themselves into pitch angles that are constant in radius and time (see Fig. 1c). The radial and azimuthal wavenumbers of the spirals match those predicted by linear theory. The values of gravitational stress  $\alpha_G$  match those of 3D simulations, and in higher resolutions we see an increase in hydrodynamical stress. Our models are significantly faster and computationally efficient compared to 3D global models of self-gravitating disks using grid and SPH methods [20,21,22].

**Related #2: Support for fragile porous dust in a gravitationally self-regulated disk around IMLup** [Ueda et al. (incl. Sudarshan) 2024, Nature Astronomy, ADS] IM Lup, a well-known T-Tauri star, is surrounded by a protoplanetary disk with spiral arms in the midplane [2,23]. We present a GI model of the IM Lup disk that offers a comprehensive explanation for diverse observations spanning from near-infrared [24,25] to millimeter wavelengths [26]. Our findings underscore the importance of dust fragility in retaining the observed millimeter emission and reveal the preference for moderately porous dust to explain observed millimeter polarization. We also find that the inner disk region is likely heated by gas accretion, providing a natural explanation for bright millimeter emission within 20 au. The actively heated

inner region in the model casts a 100-au-scale shadow, aligning seamlessly with the NIR scattered light observation.

**Main project #3: Dust dynamics in planet–disk interaction using the PLUTO code: the pressureless dust fluid method** [Ziampras, Sudarshan et al. 2025, ADS] Protoplanetary disks exhibit a rich variety of substructure in millimeter continuum emission, often attributed to unseen planets [27]. As these planets carve gaps in the gas [28], dust particles can accumulate in the resulting pressure bumps, forming bright features in the dust continuum. We investigate the role of dust dynamics in the gap-opening process with 2D radiation hydrodynamics simulations of planet–disk interaction and a two-population dust component modeled as a pressureless fluid. We consider the opacity feedback and backreaction due to drag forces as mm grains accumulate in pressure bumps at different stages of dust growth. We find that dust dynamics can significantly affect the resulting substructure (see Fig. 1d) driven by the quasi-thermal-mass planet with  $M_p/M_\star = 10^{-4}$ . We also describe and test the implementation of the publicly available pressureless dust fluid module in the PLUTO code.

**Related project #3: Streaming instability code comparison: the unstratified BA problem** [NASA TCAN Collaboration, Baronett et al. (incl. Sudarshan) in prep. [Results online] The streaming instability [29] is the most widely-accepted mechanism to grow planetesimals from mm- to cm-sized pebbles in protoplanetary disks. Several hydrodynamics codes have explored the parameters, non-linear properties, and implications of this aerodynamic instability that requires feedback between dust and gas momenta. Differences between numerical techniques (e.g., finite difference or finite volume) and dust modeling (e.g., as a pressureless fluid or as Lagrangian particles) [30,31,32] can make it difficult to disentangle unique scientific results. To address this, we present a comparison of various codes across key problems previously studied within the topic, starting with the unstratified BA problem (see Fig. 1e for the PLUTO dust fluid snapshot).

**Main Project #4: The effect of stellar irradiation on protoplanetary disk geometry: Limits to the irradiation instability with FLD** [Sudarshan et al. to be submitted] While embedded planets are a common explanation for observed substructures in protoplanetary disks, various physical processes in the disk could also trigger such features [33,34]. One such possibility that has been previously theorized for passive disks is the irradiation instability [35,36]: the flaring disk solution may become unstable [37] as directly illuminated regions puff up and cast shadows behind them. This in turn manifests as bright and dark rings in the outer regions of the disk. We investigate this problem with global axisymmetric 2.5D hydrostatic+dynamical simulations including realistic radiation transport with frequency-dependent irradiation and flux-limited diffusion. Our models showed stable, non-traveling temperature bumps within  $\sim 30$  au for models with a constant dust-to-gas mass ratio (see Fig. 1f). Additionally, we found a dependence of the shadows on the optical depth of the inner disk. We also provide a first attempt in modeling dust with a hydrostatic tracer model, showing a superheated dust irradiation layer. We highlight the role of consistently modeling the interaction of small grains on the structure of the irradiation absorption surface.

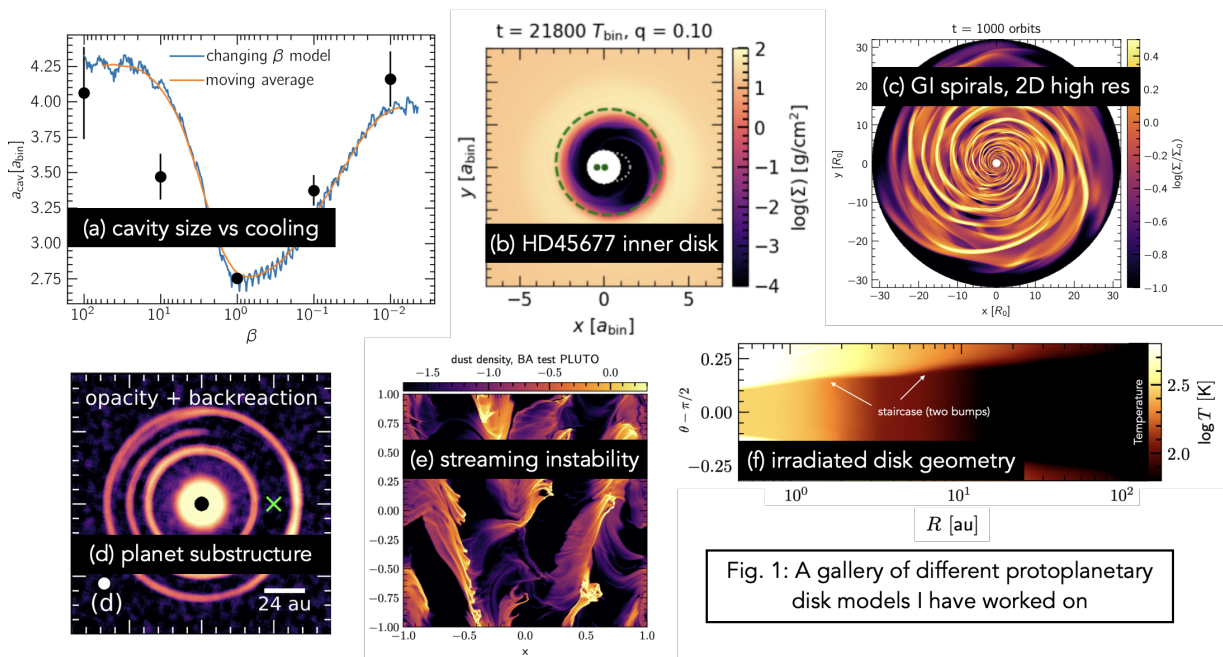


Fig. 1: A gallery of different protoplanetary disk models I have worked on

## References

1. Benisty et al. (2021)
2. Andrews et al. (2018)
3. Perotti et al. (2023)
4. Mignone et al. (2007)
5. Flock et al. (2013)
6. Ziampras et al. (2025)
7. Keppler et al. (2020)
8. Ragusa et al. (2020)
9. Thun et al. (2017)
10. Miranda & Rafikov (2020)
11. Ziampras et al. (2020)
12. Gammie (2001)
13. Lazareff et al. (2017)
14. Kluska et al. (2020)
15. Hoffmann et al. (2020)
16. Lodato (2007)
17. Kratter & Lodato (2016)
18. Meru & Bate (2012)
19. Bethune et al. (2021)
20. Rowther et al. (2020)
21. Deng et al. (2020)
22. Zier et al. (2023)
23. Huang et al. (2018)
24. Avenhaus et al. (2018)
25. Tazaki et al. (2023)
26. Sierra et al. (2021)
27. Zhang et al. (2018)
28. Rafikov (2002)
29. Youdin & Goodman (2005)
30. Benítez-Llambay et al. (2019)
31. Huang & Bai (2022)
32. Mignone et al. (2019)
33. Uribe et al. (2011)
34. Zhang et al. (2024)
35. Ueda et al. (2019)
36. Wu & Lithwick (2021)
37. Dullemond (2000)